

Vistoris



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***Close your eyes.
Would you feel safe walking through a
construction site without listening?***





How safe are construction workers, if they cannot rely on all of their senses?

What is the idea behind Visioris?

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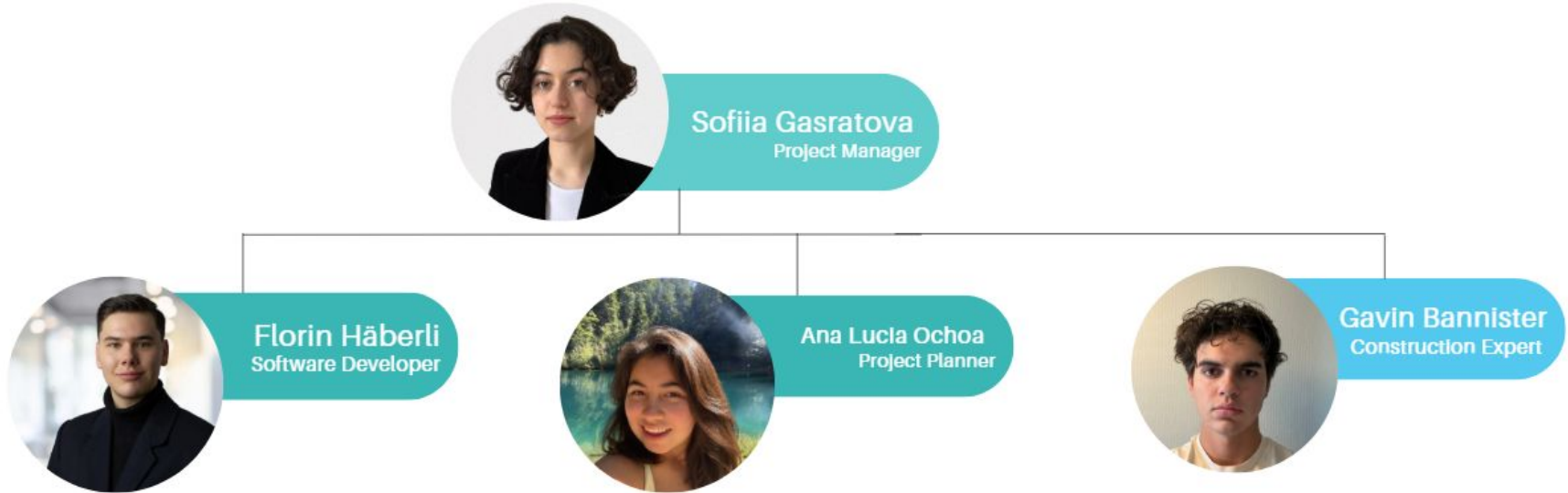
Concept: A smart, modular safety device that attaches to standard helmets.

Value Proposition: "Eyes in the back of your head"



Project Team

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Market Research

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Aspect	Market / Regulation (Abbreviation)	Key Requirement or Trend	How Vistoris Responds	Full Name of Abbreviation
Global Safety Market	PPE Market	Continuous growth in worker safety investment	Focus on accident prevention and awareness	Personal Protective Equipment
Smart Safety Technology	Smart PPE	Active safety systems using sensors and electronics	Real-time detection and vibration alerts	Smart Personal Protective Equipment
Target Industry	Construction Sector	High risk of head injuries and blind-spot accidents	Helmet-mounted rear monitoring system	—
User Expectations	End Users (Workers)	Lightweight, non-intrusive, hands-free solutions	Compact design, vibration instead of sound	—
European Regulation	EU PPE Regulation 2016/425	Accessories must not reduce protective function	Low weight, rear placement, no visual obstruction	European Union Personal Protective Equipment Regulation
United States Regulation	ANSI / ISEA Standards	Helmet accessories must not interfere with safety	Modular attachment and non-intrusive alerts	American National Standards Institute / International Safety Equipment Association
Swiss Regulation	PSA-Verordnung (PSAV)	Compliance with EU-equivalent safety rules	Designed as a complementary PPE device	Swiss Ordinance on the Safety of Personal Protective Equipment
Regulatory Principle	OSHA Guidelines	General worker safety requirements	Supports hazard awareness without distraction	Occupational Safety and Health Administration

Morphological Analysis

High-End (Green): High cost, fully certified according to regional regulations, robust

Budget (Red): Cheaper materials, limited features, no certification

Balanced (Blue): Focus on functional requirements

Selection Criteria	Option 1	Option 2	Option 3	Option 4
Attachment Method	Clip-on mount for glasses	Clip-on mount for helmet 	Elastic headband  	Magnetic plate adapter
Housing Material	ABS plastic 	Polycarbonate 	Aluminum alloy	Recycled biopolymer 
Sensor Type	Ultrasonic (HC-SR04)  	Lidar	Radar	Combined 
Sensor Placement	Back-centered 	Dual side sensors	Sides & Back 	Detachable modules 
Feedback Type	Haptic vibration motor 	Audio beep	LED indicator 	Combined 
Power Source	Rechargeable Li-ion battery  	Replaceable AAA battery 	External helmet power	Solar-assisted charging
Microcontroller Platform	ESP32  	RP2040	SM32 	Arduino Nano
Weather Protection	IP52 (basic dust/splash) 	IP65 (industrial)  	IP67 (waterproof)	IP68 (fully sealed)
User Interface	Single power button  	Mobile app control 	Touch Panel	Voice activation
Alert Directionality	General warning  	Directional (left/right)	3D spacial cue	Adjustable sensitivity zones 
Development Environment	Arduino IDE 	Platform IO  	-	-
Safety Standard	European	Swiss 	American (United States) 	- 

Product Variants & Decision Process

Value-Benefit Analysis

Criteria	Weight	Variant 1 - High-end		Variant 2 - Budget		Variant 3 - Balanced	
		Score	Weighting	Score	Weighting	Score	Weighting
Collision detection	15	4	60	2	30	3	45
Collision alerting	15	5	75	1	15	2	30
Usability	15	4	60	3	45	4	60
Battery Life	10	3	30	4	40	4	40
Weight	10	1	10	3	30	3	30
Durability	10	3	30	2	20	2	20
Production Cost	10	4	40	5	50	3	30
Affordability	5	1	5	4	20	2	30
Expandability	5	5	25	2	10	3	15
Customizability	5	4	20	1	5	3	15
Total	100	355		265		315	

Variant Decision: **Balanced**

- Proves **concept** and **functionality** without burden of immediate certification

Technical Specification

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Hardware Specs:

1. **Core:** ESP32 Microcontroller.
2. **Sensors:** 3x Proximity/Motion sensors covering 1.5m – 4.5m range.
3. **Feedback:** 3x Vibration motors for directional alerting.
4. **Power:** Rechargeable Li-Ion battery (Target: ≥ 20 h active use).



Design:

1. **Housing:** ABS Plastic (durable, lightweight).
2. **Mounting:** (fits standard construction helmets).
3. **Response Time:** < 100 ms (Critical for safety).



- **Purpose**

Ensure the system design meets specifications by theoretical evaluation of the concept

- **Key Areas Assessed**

- **Sensors & detection**
- **Robustness**
- **Usability**

- **Outcome**

- Verified design principles
- Deemed feasible for further development



- **Purpose**

Assess user perception and device's feedback

- **Key Areas Assessed**

- **Signal interpretation**
- **System helpfulness**
- **User satisfaction**

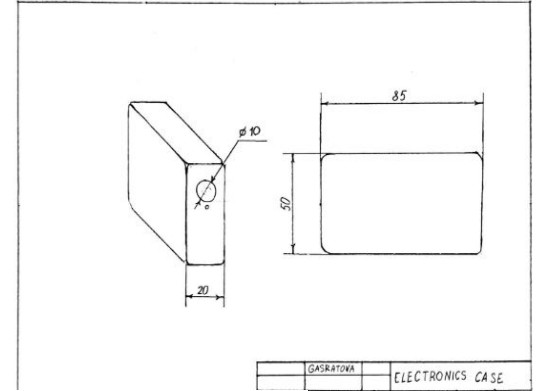
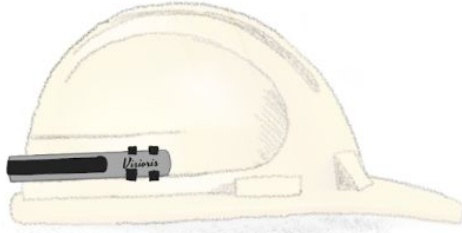
- **Outcome**

The system provides meaningful assistance in improving situational awareness

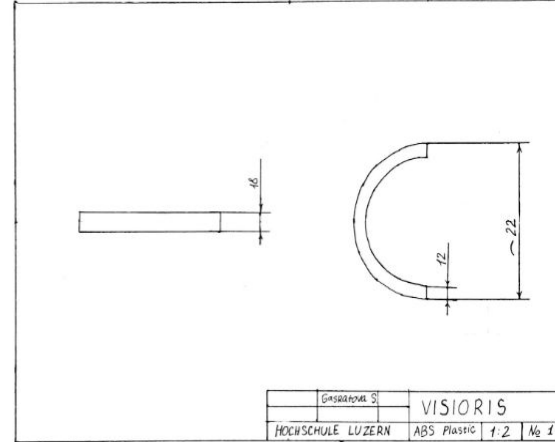


Construction Plans

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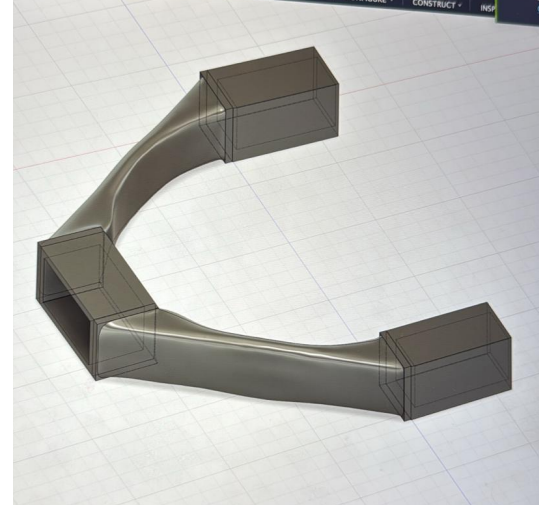
GASKETOWN	ELECTRONICS CASE
HLE LUZERN	ABS Plastic 1.1 No 2



GASKETOWN S	VISTORIS
HOCHSCHULE LUZERN	ABS Plastic 1.2 No 1

Mockup Conception

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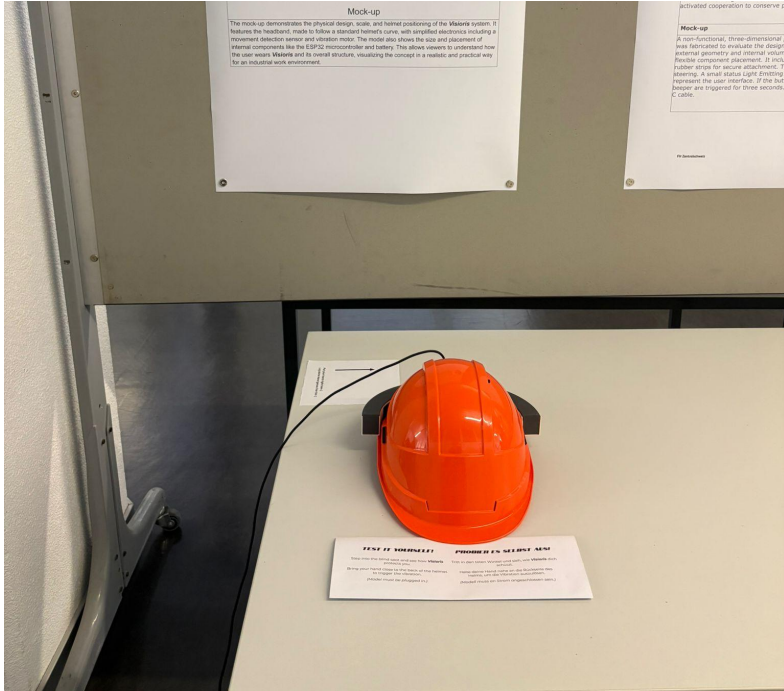
Mockup Construction

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Mockup Demonstration

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Conclusion

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Thank you!

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